

Point-only input

```
{  
  "requested_ks": [32, 64, 128],  
  "n_points": 1024,  
  "leave_self_out": true  
}
```

Leave-self-out neighborhoods

Query each point in a spatial index; exclude its row.
 $k \in \{32, 64, 128\}$, capped at $N - 1$; $N \geq 9$.

configuration illustration

$k = 32$ **neighbors**

Own PCA frame
Quadratic surface fit

$k = 64$ **neighbors**

Own PCA frame
Quadratic surface fit

$k = 128$ **neighbors**

Own PCA frame
Quadratic surface fit

$$\delta_k = c_{0,k} n_k$$

Project & combine corrections

Project δ_k onto the $k = 64$ normal.
$$d = \sum_k w_k (\delta_k \cdot n_{64})$$

Weights from inverse prediction-variance proxies.

Continuous uncertainty shrinkage

$$\alpha = \frac{v_n}{v_n + v_p + v_s + \varepsilon}$$

Residual variance, prediction variance,
and 3-D scale disagreement.

Input-only diagnostics

weights, alpha
noise_variance, prediction_variance
scale_disagreement

Heuristic proxies; no calibrated safety guarantee.

Normal-only update

$$p' = p + \alpha d n_{64}$$

multiscale_consensus

Same point count and source-row order.

Independent evaluation after outputs are sealed

Independent DTU reference

Used only for evaluation; never fed into fitting.

Fixed-region evaluation

Point-to-reference MAE, recall and F-score.